

Differentiation by first principles



1 Consider the function $f(x) = x^3$.

- a Complete the limiting chord process for $f(x) = x^3$ at the point $x = 3$ by filling in the table of values. Round your answers to four decimal places.

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
3	4	1	
3	3.5		
3	3.1		
3	3.05		
3	3.01		
3	3.001		
3	3.0001		

- b Use the limiting chord process to calculate the instantaneous rate of change for the remaining values of x in the table:

x	1	2	3	4	5
Instantaneous rate of change $f'(x)$ at x	3	12		48	

- c Find the instantaneous rate of change of $f(x)$ at any point x .

2 Consider the function $f(x) = x^2$.

- a Complete the limiting chord process for $f(x) = x^2$ at the point $x = 3$ by filling in the table of values:

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
3	4	1	
3	3.5		
3	3.1		
3	3.05		
3	3.01		
3	3.001		
3	3.0001		

- b Use the limiting chord process to calculate the instantaneous rate of change for the remaining values of x in the table:



x	1	2	3	4	5
Instantaneous rate of change $f'(x)$ at x	2	4		8	

- c Find the instantaneous rate of change of $f(x)$ at any point x .

3 Consider the function $f(x) = 2x^2$.

- a Complete the limiting chord process for $f(x) = 2x^2$ at the point $x = 3$ by filling in the table of values. Round your answers to four decimal places.

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
3	4	1	
3	3.5		
3	3.1		
3	3.05		
3	3.01		
3	3.001		
3	3.0001		

- b Use the limiting chord process to calculate the instantaneous rate of change for the remaining values of x in the table:

x	1	2	3	4	5
Instantaneous rate of change of $f(x)$ at x	4	8		16	

- c Hence, find the instantaneous rate of change of $f(x)$ for any given value of x .

4 For each of the following functions:



i Find $f(x+h)$.

ii Find $f(x+h) - f(x)$.

iii Find $\frac{f(x+h) - f(x)}{h}$.

a $f(x) = 3x + 2$

b $f(x) = 8 - x$

c $f(x) = 2x - 3$

d $f(x) = -5x + 4$

e $f(x) = 5 - x^2$

f $f(x) = x^2 + 3x + 6$

g $f(x) = \frac{1}{x}$

h $f(x) = \frac{1}{7x}$

i $f(x) = 3x^2 - 4x + 2$

5 For each of the following functions:

i Find $f(x+h)$ in expanded form.

ii Find $f(x+h) - f(x)$.

iii Find $\frac{f(x+h) - f(x)}{h}$.

iv Hence, find $f'(x)$ by evaluating $\lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right)$.

a $f(x) = x^2$

b $f(x) = 4x^2$

c $f(x) = -6x^2$

6 Find the derivative of each of the following functions from first principles:

a $f(x) = 6$

b $f(x) = -6$

c $f(x) = 5x$

d $f(x) = -2x$

e $f(x) = 2x + 3$

f $f(x) = x^2 - 3$

g $f(x) = 4x^2 + 5$

h $f(x) = x^2 - 3x$

i $f(x) = 2x^2 - 3x$

j $f(x) = 4x^2 - 3x - 5$

k $f(x) = \sqrt{x}$

l $f(x) = \sqrt{x-5}$



Gradients

- 7 Consider the function $f(x) = 4x$.
- Is the gradient of the function constant or variable?
 - Find the gradient of the function using the points $(1, 4)$ and $(7, 28)$ that lie on the function.
 - Find the gradient of the function from first principles.
- 8 Consider the function $f(x) = (x + 6)(x + 2)$.
- Find the gradient function of $f(x) = (x + 6)(x + 2)$ from first principles.
 - Find the gradient at the point where $x = 3$.
- 9 Consider the function $y = \sqrt{x}$.
- Find the gradient function of $f(x) = \sqrt{x}$ from first principles.
 - Find the gradient at the point where $x = 25$.
- 10 Consider the function $f(x) = 7 - x^2$.
- Find the gradient function $f'(x)$ using first principles.
 - Hence, calculate the gradient of $f(x)$ at the point where $x = -1$.
- 11 Consider the function $f(x) = x^2 + 3x + 4$.
- Find the gradient function $f'(x)$ using first principles.
 - Hence, calculate the gradient of $f(x)$ at the point where $x = 4$.
- 12 Consider the function $f(x) = x^3$.
- Find the gradient function $f'(x)$ using first principles.
 - Hence, calculate the gradient of $f(x)$ at the point where $x = 2$.
- 13 Consider the function $f(x) = x^4$.
- Find the gradient function $f'(x)$ using first principles.
 - Hence, calculate the gradient of $f(x)$ at the point where $x = -5$.

14 Find the gradient of $f(x) = 3x^2$ at the point  (48) from first principles.

15 Find the gradient of $f(x) = 3x^3$ at the point $(-4, -192)$ from first principles.

16 Find the gradient of $f(x) = 2x(x - 6)$ at the point $(2, -16)$ from first principles.

17 Find the gradient of $f(x) = 3x(x - 2)^2$ at the point $(4, 48)$ from first principles.

18 Consider the function $y = \frac{1}{x}$.

a Find the gradient function of $y = \frac{1}{x}$ from first principles.

b Find the x -values of the two points on the function at which the gradient is $-\frac{1}{16}$.

19 Consider the function $f(x) = \frac{1}{2x - 4}$.

a Find $f'(x)$ from first principles.

b Hence, find $f'(4)$.